

Mrittika Nandi

PHD (CRYPTOGRAPHY RESEARCH LAB)
DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
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EDUCATION

Indian Institute of Science Education and Research(IISER), Bhopal, India
PhD (Lattice-Based Cryptography)
Electrical Engineering and Computer Science *Jan' 26 - Present*

Indian Association for the Cultivation of Science, Jadavpur, Kolkata, India
Master of Science, School of Mathematical and Computational Sciences with Mathematics as major *Sept' 21 - July' 23*
CGPA: 7.85/10

Basanti Devi College, University of Calcutta, Kolkata, India
Bachelor of Science, Mathematics *Jul' 18 - Aug' 21*
CGPA: 8.822/10

South Point High School (CBSE), Kolkata, India
Higher Secondary *Apr' 16 - May' 18*
Percentage: 86.2

South Point High School (WBBSE), Kolkata, India
Secondary *Apr' 03 - Mar' 16*
Percentage: 90.57

RESEARCH INTERESTS

Lattice-based cryptography: Exploring post-quantum secure primitives based on hard lattice problems, with applications to digital signatures, blind signatures, identity-based and searchable encryption, secure multi-party computation, secret sharing, privacy-preserving protocols, and blockchain systems.

Zero-knowledge proofs: In both pre-quantum and post-quantum setting, zero-knowledge proofs have immense applications especially in fields where anonymity are of primary importance. I am interested to use ZK proofs (NIZK/zk-SNARKS) to reduce the overhead of lattice-based approaches to make them lightweight and practical.

Lattice Reduction: The current state-of-the-art lattice basis reduction algorithms and their applications in the cryptanalysis of lattice-based protocols fascinate me. I am interested to work on making the enumeration and sieving algorithms for short vector generation more efficient and also use them as the SVP oracle in the BKZ algorithm.

PROJECTS

New designs of post-quantum cryptographic candidates
Project JRF (DRDO), *Supervisor: Dr. Shashank Singh* *Mar'24-Dec'25*

- Thoroughly cryptanalyzed Dilithium which includes understanding the parameter choices based on lattice reduction attacks and Core-SVP estimation techniques.
- Designed a new lattice-based digital signature scheme with the same hardness assumptions as Dilithium but with faster signing (less rejection/repetition).
- Simulated the behaviour of a BKZ reduced basis using the ZGSA assumption to obtain the optimal parameters for my scheme
- Implemented the scheme in python and benchmarked its performance for the NIST suggested security levels.

DPI over encrypted traffic

SparQ Summer Internship, *QNu Labs*

May'25 - Aug'25

- Investigated how middleboxes can accomplish deep-packet-inspection (DPI) on encrypted HTTPS traffic.
- Studied the 2015 paper “BlindBox: DPI over encrypted traffic”.
- Gained a deep understanding of practical techniques for performing secure DPI.

Wiener-Ito Integrals

MS Project, *Supervisor: Prof. Alok Goswami*

Sept'22-May'23

- Studied Brownian Motion, Markov Processes, Wiener and Ito Integrals
- Applied the Ito formula to probabilistically solve the Dirichlet Problem, Heat Equation and some other parabolic PDEs

Flight Delay Prediction using Logistic Model

Summer Project, *Supervisor : Prof. Kiranmoy Das*

July'22-Sept'22

- Gained expertise in running various machine-learning algorithms like k-means algorithm on a dataset from Kaggle along with hands-on experience in programming with R..
- Learned to use linear and logistic regression models for better predictive performance.

RELEVANT COURSES

Relevant Courses (Audit)

- Modern Cryptography, Advanced Algorithms, Algebraic Number Theory
- Module Theory, Lattices and hard problems on lattices, Lattice Basis Reduction (LLL,BKZ), Lattice-based protocols, Zero-knowledge proofs (Research Work at IISER Bhopal)

Postgraduate Courses

- Algebra (Field Theory, Galois Theory)
- Object-Oriented Programming with C++
- Measure Theory and Probability
- Statistics with R
- MS Project (Brownian Motion, Stochastic Processes, Stochastic Integrals)

Undergraduate Courses

- Abstract Algebra (Group Theory, Ring Theory)
- Linear Algebra
- Classical Algebra (Number Theory)
- Probability and Statistics
- Discrete Mathematics
- Data Structure and Algorithm
- Programming (C and Python)

PUBLICATIONS

Arup Mazumder, **Mrittika Nandi**, Shashank Singh (2026). “Round-Optimal Identity-Based Blind Signature from Module Lattice Assumptions”. IACR ePrint arch. [MNS26]

Ayan Dutta, **Mrittika Nandi**, Smita Sarkar, Swetlina Hota, Suklav Ghosh (2023). “A MODEL-BASED APPROACH FOR FLIGHT DELAY PREDICTION”. Indian Journal Of Applied Research, 13(5), 36–40. [DNS⁺23]

OTHER SCHOLASTIC ACHIEVEMENTS

Qualified UGC NET June 2025 in the PhD only category

Secured highest marks in Mathematics and held the top rank across all science disciplines during my undergraduate studies (B.Sc.)

Q&A Expert in Advanced Mathematics at Chegg India

EXPERIENCE

Attended the “Next-Gen Cybersecurity Workshop: Preparing for the Post-Quantum Era (Online Mode)” by IIT Indore in July, 2025.

Attended NIWC 2024 workshop on Lattice-based Cryptography held at MNNIT, Allahabad

Attended a talk on Recent Advances in Cryptology by Dr. Bimal Roy at IACS, Kolkata

Attended a Webinar on Cryptography and Web Security on 26/08/20 at Lady Brabourne College

LANGUAGES

Bengali, English,Hindi

SKILLS

Python, Sage, C, C++, R

REFERENCES

Dr. Shashank Singh
Assistant Professor, EECS, IISER Bhopal
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Dr. Amit Kumar Chauhan

Senior Research Associate, QNu Labs Pvt. Ltd., Bangalore, India

E-mail: amit.c@qnulabs.com

Prof. Alok Goswami (Retired Professor, ISI Kolkata)

Visiting Professor, School of Mathematical and Computational Sciences, IACS Kolkata

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Prof. Kiranmoy Das

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